

Semi-autonomous robotic system that interfaces human kinematics

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Introduction and System Architecture

1. Introduction

This project details the design and implementation of a wireless, hand-gesture-controlled robotic rover. The system translates physical hand movements into directional commands using an MPU6050 inertial measurement unit (IMU). To ensure robust telemetry, the architecture utilizes the ESP-NOW protocol between two ESP32 microcontrollers, bypassing traditional Wi-Fi network overhead. Furthermore, the rover integrates a hardware-level safety veto system using ultrasonic and infrared sensors to prevent collisions and drops, alongside a custom analog battery monitoring circuit to protect the power supply.

2. Objective

Traditional remote-controlled ground vehicles rely on tactile interfaces (joysticks or buttons). This project proposes a more intuitive human-machine interface utilizing spatial gesture recognition. The primary objectives of this project were to:

- Successfully map 3D spatial orientations (pitch and roll) to 2D kinematic movements.
- Establish a low-latency, peer-to-peer wireless communication link.
- Implement semi-autonomous hardware overrides to protect the rover from environmental hazards (walls and cliffs).
- Design an analog voltage comparator circuit to monitor a multi-cell lithium-ion power source.

3. System Architecture

The system is divided into two discrete nodes: the Transmitter and the Receiver (Rover).

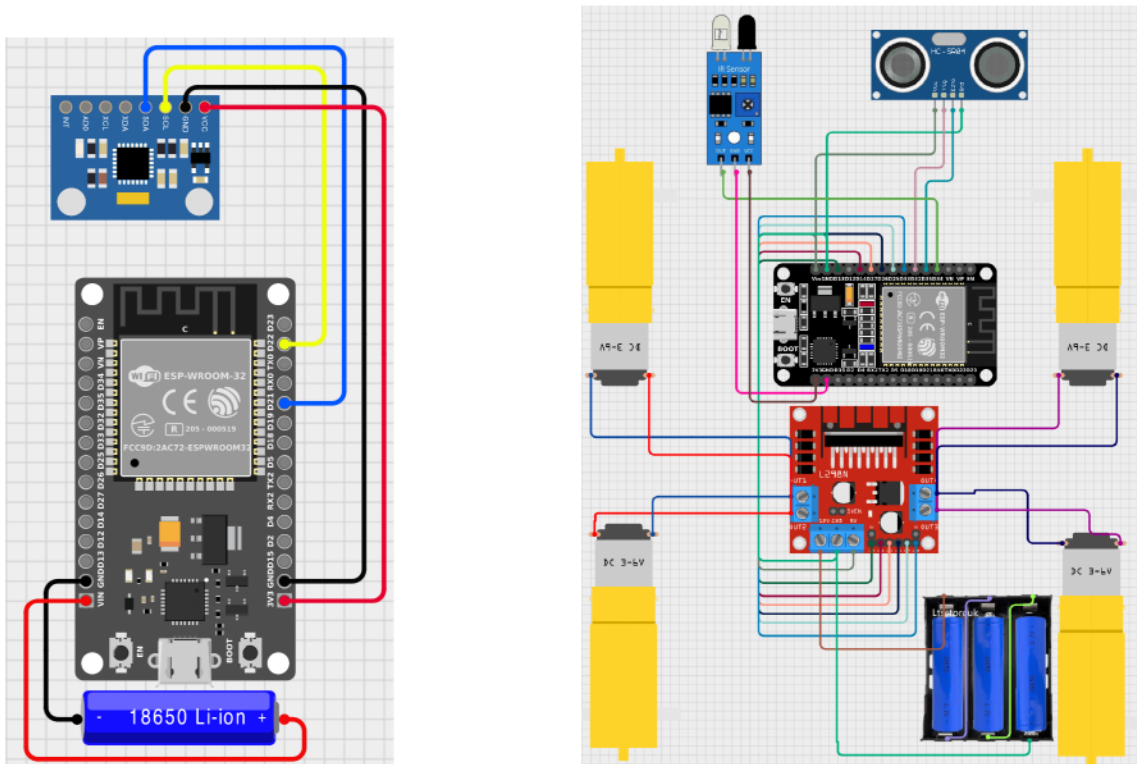
3.1 The Transmitter Node

The wearable node acts as the system's brain. An ESP32 microcontroller interfaces with an MPU6050 via the I2C protocol. The MPU6050 continuously captures accelerometer and gyroscope data. The MPU6050 determines pitch and roll using its internal accelerometer to measure Earth's gravity across its X, Y, and Z axes. The microcontroller takes these axial force readings and applies trigonometric functions (specifically the arctangent) to calculate the exact angle of tilt relative to the ground. To ensure smooth and accurate readings, this gravity data is typically fused with the module's gyroscope data using a mathematical algorithm (like a complementary filter) to eliminate vibration noise and sensor drift. The ESP32 processes this raw data, calculates the pitch and roll angles, and categorizes them into discrete command states (Forward, Backward, Left, Right, Stop). These commands are transmitted wirelessly using the ESP-NOW protocol directly to the receiver's MAC address.

3.2 The Receiver Node (Rover Chassis)

The mobile node consists of a second ESP32 microcontroller, an L298N dual H-bridge motor driver controlling four DC motors, and a sensor array. Upon receiving a command string, the ESP32 evaluates local environmental

data (from the ultrasonic and IR sensors) before actuating the L298N driver.



Hardware Implementation & Safety Mechanisms

4. Hardware Components

- **Microcontrollers:** 2x ESP32 Development Boards
- **Sensors:** MPU6050 (IMU), HC-SR04 (Ultrasonic), IR Obstacle Avoidance Module
- **Actuation:** L298N Motor Driver, 4x DC Gear Motors
- **Power:** 3S 18650 Lithium-Ion Battery Pack (12.6V max)
- **Analog Circuitry:** LM358 Operational Amplifier, Zener Diode, Resistors, LED

5. The Autonomous Safety Veto System

To address the inherent latency in human reaction time, the rover features a built-in safety override system that operates independently of the user's commands.

- **Frontal Collision Prevention:** An HC-SR04 ultrasonic sensor is mounted on the front chassis. The ESP32 pulses the trigger pin and calculates the echo duration to measure distance. If an obstacle is detected within 15 cm, the microcontroller immediately revokes any active "Forward" command.
- **Edge/Drop Detection:** An Infrared (IR) sensor is mounted at the front-center of the chassis, angled downward. If the surface disappears (indicating a ledge or staircase), the lack of IR reflection triggers a digital LOW signal, instantly halting forward momentum.
- **Active Momentum Cancellation:** To rapidly counteract the rover's physical inertia, the system executes a momentary maximum-power reverse thrust (75ms) before engaging the motor locks. This active reversal neutralizes forward momentum, drastically reducing stopping distance to prevent collisions and edge drops more effectively than standard electronic braking.

6. Power Management and Analog Monitoring

The system is powered by a 3-cell (3S) 18650 battery pack providing a peak voltage of 12.6V, which is stepped

down by the L298N's internal regulator to safely power the 3.3V logic of the ESP32.

To prevent catastrophic over-discharge of the lithium-ion cells, a custom battery monitor was designed using an LM358 operational amplifier configured as an analog voltage comparator. A Zener diode establishes a stable reference voltage on the non-inverting input. A precision voltage divider scales the battery's fluctuating voltage and feeds it into the inverting input. When the pack voltage drops below the critical 10.0V threshold, the comparator output swings HIGH, illuminating a warning LED to alert the user.

Communication Protocol and Conclusion

7. Wireless Protocol: The ESP-NOW Advantage

Early iterations of this project utilized **nRF24L01 radio modules** communicating over the SPI bus. However, managing the high inrush currents of the PA+LNA amplifiers alongside the noisy power delivery of the motor driver resulted in frequent SPI bus brownouts and hardware-level packet loss.

To eliminate these hardware failure points, the architecture was successfully migrated to **ESP-NOW**. ESP-NOW is a connectionless, peer-to-peer wireless protocol developed by Espressif that operates directly at the Wi-Fi MAC layer. Instead of connecting to a central router or spending time performing standard Wi-Fi handshakes, devices using ESP-NOW send data packets directly to one another using their unique hardware MAC addresses. By bypassing this network overhead, it creates a highly robust, ultra-low-latency communication link that is perfect for instantaneous, real-time motor control. This transition provided three distinct advantages:

Hardware Reduction: Eliminated external SPI wiring, dedicated 3.3V regulators, and physical antennas.

Latency: Achieved near-instantaneous transmission (sub-millisecond) without the overhead of standard Wi-Fi handshakes.

Reliability: Provided a highly robust, interference-resistant link crucial for continuous real-time motor control.

8. Challenges and Troubleshooting

The primary challenge involved power starvation during peak loads. Initial testing revealed that the high current draw from the motors caused transient voltage drops, resetting the microcontrollers and severing the telemetry link. This was resolved by implementing strict power domain isolation (separating logic power from motor power) and moving to the native ESP-NOW protocol, which proved significantly more resilient to power fluctuations than external SPI peripherals.

9. Conclusion

The project successfully demonstrates a highly responsive, gesture-controlled robotic platform. The integration of ESP-NOW ensures reliable telemetry, while the hardware-level safety overrides successfully protect the rover from user error or environmental hazards.

10. TinyML

Regarding the TinyML integration, I initially attempted to deploy a custom machine learning model on the Grove ESP32 using Edge Impulse. I collected extensive MPU6050 telemetry to train a neural network to recognize complex, continuous hand gestures rather than relying on simple tilt angles. However, during live testing, the deployed model yielded a very low classification accuracy and struggled to consistently distinguish between overlapping spatial movements. The inference results were too erratic for reliable, real-time motor control, frequently resulting in false positives or missed drive commands. To guarantee the rover's operational stability and immediate responsiveness, I decided to exclude the ML model from the current build and revert to the highly reliable pitch/roll threshold logic.